Individual Reflection

Advanced Algorithms Assignment 2
Zeeshan Ahmed [25494269]

The worst-case for Graham Scan is (O(nlog n)), primarily due to the initial sorting of points by their polar angle relative to the pivot. Although the algorithm has an (O(nlog n)) time complexity regardless of the input distribution (algorithm is not output sensitive), certain inputs can introduce additional challenges during implementation. For example, datasets containing many collinear points or points sharing the same polar angle require careful handling to ensure that the resulting hull contains only the appropriate boundary points. Incorrect handling of these cases can lead to redundant points or an invalid convex hull.

One of the more challenging aspects of my implementation was determining how to efficiently compare point orientations during the sorting and scanning stages. Initially, I relied on AI-generated code to assist with the implementation. Despite explicitly specifying that I wanted to use the cross product rather than atan2 for determining point orientation, the generated implementation still incorporated atan2 for angle calculations. I accepted this implementation without sufficiently questioning the design decision. This introduced unnecessary computational overhead, as calculating inverse trigonometric functions for every point during sorting is more expensive than relying on cross-product comparisons. I eventually recognised that I needed a stronger understanding of the underlying mathematics rather than relying on generated code without fully evaluating its implementation choices.

If I were to implement Graham Scan again from scratch, I would be more intentional about the algorithm's design and avoid introducing unnecessary operations such as atan2. Instead, I would prioritise taking greater care when reviewing AI-generated suggestions, ensuring that they align with the algorithm's theoretical foundations and my original implementation requirements.

One thing I did not fully appreciate before implementing Graham Scan was the computational overhead associated with seemingly straightforward mathematical operations. I initially understood that sorting points by polar angle was necessary, but I had not considered how much additional work could be introduced by calculating the actual angles using atan2. I now better understand that the cross product provides sufficient information to compare orientations

without needing to calculate explicit angles. This distinction helped me appreciate the importance of selecting appropriate mathematical operations when implementing algorithms, particularly when those operations are executed repeatedly across large datasets.